

IT2120 - Probability and Statistics

Lab Sheet 10

IT24103586 – Herath H.M.T.P

Exercise

Instructions: Create a folder in your desktop with your registration number (Eg: "IT....."). You need to save the R script file and take screenshots of the command prompt with answers and save it in a word document inside the folder. Save both R script file and word document with your registration number (Eg: "IT....."). After you finish the exercise, zip the folder and upload the zip file to the submission link.

1. A vending machine owner claims that customers choose the four snack types (A, B, C, D) with equal probability. To test this claim, a researcher records the number of purchases for each snack type during one week and results are given below.

Snack_Type	Count
A	120
B	95
C	85
D	100

- i. State the null and alternative hypotheses for the test.
- ii. Perform a suitable chi-squared test to test the null hypothesis.
- iii. Give your conclusions based on the results.

i)

- **Null Hypothesis (H_0):**

Customers choose the four snack types (A, B, C, and D) **with equal probability**.

$$p_A = p_B = p_C = p_D = 0.25$$

- **Alternative Hypothesis (H_1):**

Customers **do not** choose the four snack types with equal probability.

ii)

```
IT24103586_Lab_10.R* x
Source on Save
1 setwd("C:\\Users\\thisu\\OneDrive\\Desktop\\IT24103586")
2
3 #Observed frequencies
4 snack_counts <- c(A = 120, B = 95, C = 85, D = 100)
5 snack_counts
6
7 # Expected probabilities (equal probability = 0.25 each)
8 expected_prob <- rep(1/4, 4)
9
10 # Perform Chi-Square Goodness-of-Fit Test
11 chi_result <- chisq.test(snack_counts, p = expected_prob)
12
13 # Display test results
14 print(chi_result)
15
16 # Display expected frequencies (to check assumptions)
17 chi_result$expected
18

20:33 (Top Level) ↕

Console Terminal x Background Jobs x
R 4.5.1 · C:/Users/thisu/OneDrive/Desktop/IT24103586/ ↗
> #Observed frequencies
> snack_counts <- c(A = 120, B = 95, C = 85, D = 100)
> snack_counts
  A   B   C   D
120 95  85 100
>
> # Expected probabilities (equal probability = 0.25 each)
> expected_prob <- rep(1/4, 4)
>
> # Perform Chi-Square Goodness-of-Fit Test
> chi_result <- chisq.test(snack_counts, p = expected_prob)
>
> # Display test results
> print(chi_result)

      Chi-squared test for given probabilities

data:  snack_counts
X-squared = 6.5, df = 3, p-value = 0.08966

>
> # Display expected frequencies (to check assumptions)
> chi_result$expected
  A   B   C   D
100 100 100 100
> |
```

iii)

```
19 # Conclusion based on p-value
20 if (chi_result$p.value < 0.05) {
21   cat("\nConclusion: Since p-value <", 0.05, ", reject H0.\n")
22   cat("There is a significant difference in the proportion of snack choices.\n")
23 } else {
24   cat("\nConclusion: Since p-value >=", 0.05, ", fail to reject H0.\n")
25   cat("There is no significant difference – snack choices are equally likely.\n")
26 }
27
```

27:1 (Top Level) ↕

Console **Terminal** ✕ **Background Jobs** ✕

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```
> # Conclusion based on p-value
> if (chi_result$p.value < 0.05) {
+   cat("\nConclusion: Since p-value <", 0.05, ", reject H0.\n")
+   cat("There is a significant difference in the proportion of snack choices.\n")
+ } else {
+   cat("\nConclusion: Since p-value >=", 0.05, ", fail to reject H0.\n")
+   cat("There is no significant difference – snack choices are equally likely.\n")
+ }
```

Conclusion: Since p-value >= 0.05 , fail to reject H0.
There is no significant difference – snack choices are equally likely.
> |

- Since p-value = 0.2212 > 0.05, Fail to reject H_0 .